Box-Cox gamma sweep | Val ANN10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0689	0.279	0.913	0.853	0.282	1	0.177	0.935	1.08	0	18/18
1 ANN10	0.1	0.0473	0.249	0.958	0.838	0.251	2.81	0.194	0.982	1.19	0	18/18
1 ANN10	0.2	0.0451	0.358	0.945	0.949	0.314	0.973	0.125	0.97	1.04	0	18/18
1 ANN10	0.3	0.0283	0.276	0.667	0.895	0.141	0.898	0.0493	1.42	12.7	0	18/18
1 ANN10	0.5	0.0606	0.584	0.965	0.999	0.175	0.918	0.0938	1	1	0	18/18
1 ANN10	1	0.193	0.903	1.56	1.12	0.801	0.98	0.386	1.17	52.4	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep | Val P-only DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
2 P-only DeepONet10	0.05	0.182	0.779	0.955	0.991	0.267	2.76	0.28	0.998	1	0	18/18
2 P-only DeepONet10	0.1	0.0508	0.0899	0.932	0.966	0.0963	0.953	0.0673	0.972	0.99	0	18/18
2 P-only DeepONet10	0.2	0.0606	0.379	0.884	0.946	0.092	0.615	0.104	1.03	1.38	0	18/18
2 P-only DeepONet10	0.3	0.235	0.514	0.957	10.3	0.236	2.24	0.386	1.14	11.1	0	18/18
2 P-only DeepONet10	0.5	0.324	0.808	2.2	2.4	0.83	1.56	0.443	0.984	3.15	0	18/18
2 P-only DeepONet10	1	0.179	0.637	1.37	2.09	0.698	0.985	0.273	1.07	22.5	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep | Val ANN10 + P-only DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
3 ANN10 + P-only DeepONet10	0.05	0.1	0.508	0.952	0.938	0.223	1.96	0.231	0.998	1.06	0	18/18
3 ANN10 + P-only DeepONet10	0.1	0.035	0.0815	0.818	0.966	0.0525	0.595	0.0485	0.662	1.01	0	18/18
3 ANN10 + P-only DeepONet10	0.2	0.036	0.145	0.864	0.737	0.106	0.213	0.0667	0.971	1.23	0	18/18
3 ANN10 + P-only DeepONet10	0.3	0.0433	0.445	0.843	0.871	0.145	1.04	0.0843	1.03	1.83	0	18/18
3 ANN10 + P-only DeepONet10	0.5	0.0597	0.525	1.47	1.18	0.437	1.82	0.113	0.958	16.1	0	18/18
3 ANN10 + P-only DeepONet10	1	0.0874	0.593	1.16	1.37	0.6	0.762	0.216	1.32	37.5	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep | Val T/P DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 24, 33, 41, 46, 63, 71, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
4 T/P DeepONet10	0.05	0.222	0.656	1	1	0.67	1	0.247	1.26	1.5	0	18/18
4 T/P DeepONet10	0.1	0.0711	0.48	1	0.813	0.124	0.957	0.108	0.999	0.999	0	18/18
4 T/P DeepONet10	0.2	0.106	0.768	0.709	1	0.319	0.396	0.168	1	1	0	18/18
4 T/P DeepONet10	0.3	0.0827	0.298	0.899	0.971	0.161	0.974	0.214	0.999	1	0	18/18
4 T/P DeepONet10	0.5	0.52	0.7	3.17	8.27	0.687	2.29	0.309	26.7	28.9	0	13/18
4 T/P DeepONet10	1	0.236	2.57	1	1.82	0.988	0.549	0.36	0.998	30.8	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep | Val ANN10 + T/P DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
5 ANN10 + T/P DeepONet10	0.05	0.195	0.332	1.68	0.981	0.591	2.49	0.222	1.59	1.59	0	18/18
5 ANN10 + T/P DeepONet10	0.1	0.013	0.0698	0.739	0.598	0.0845	0.899	0.0764	0.842	0.882	0	18/18
5 ANN10 + T/P DeepONet10	0.2	0.0241	0.255	0.444	0.673	0.169	0.18	0.0741	0.986	1	0	18/18
5 ANN10 + T/P DeepONet10	0.3	9.41e-03	0.115	0.312	0.326	0.0766	0.158	0.0215	0.812	0.895	0	18/18
5 ANN10 + T/P DeepONet10	0.5	0.0918	0.804	1.04	2.57	0.734	1.12	0.185	1.18	1.76	0	18/18
5 ANN10 + T/P DeepONet10	1	0.157	0.738	2.26	1.77	0.822	0.578	0.282	0.983	27.1	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep | Val T/P/ϕ DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
6 T/P/ϕ DeepONet10	0.05	0.421	0.415	5.08	0.991	0.374	4.6	0.356	4.12	2.62	0	18/18
6 T/P/ϕ DeepONet10	0.1	0.117	0.439	1	0.88	0.252	1	0.11	1	2.8	0	18/18
6 T/P/ϕ DeepONet10	0.2	0.0127	0.171	0.35	0.546	0.234	0.299	0.0399	0.839	0.803	0	18/18
6 T/P/ϕ DeepONet10	0.3	0.345	0.67	1	1.27	0.414	1.28	0.324	1.31	2.53	0	18/18
6 T/P/ϕ DeepONet10	0.5	0.207	0.415	1.05	1	0.83	1	0.281	1	1	0	17/18
6 T/P/ϕ DeepONet10	1	0.125	0.967	2.52	1.91	0.878	0.467	0.179	1.15	10.8	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.

Box-Cox gamma sweep | Val ANN10 + T/P/ϕ DeepONet10: common ICs [0, 9, 10, 14, 17, 19, 23, 24, 29, 33, 41, 46, 48, 52, 63, 71, 75, 81]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.142	0.385	0.988	0.977	0.663	1	0.378	1.29	2.38	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0442	0.103	0.724	0.345	0.105	0.942	0.0906	0.842	0.865	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.013	0.138	0.369	0.285	0.0847	0.268	0.0388	0.808	0.929	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0751	0.361	0.988	0.993	0.153	0.926	0.141	1	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0582	0.678	0.854	0.866	0.631	0.757	0.173	0.981	0.974	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	1	0.186	0.989	2.76	1.24	0.848	0.468	0.211	1.11	15.3	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.